

PROJECT ABSTRACT

RISC-V Based SoC with Hardware CNN Accelerator for Handwritten Digit Recognition

1. Problem Statement

Modern embedded systems increasingly require image-processing and machine-learning capabilities while operating with limited computational and memory resources. Handwritten digit recognition is a compact and meaningful image-classification application that requires a significant number of arithmetic operations, particularly multiplication and accumulation operations used in Convolutional Neural Networks (CNNs). Executing these operations entirely in software can place a high computational load on a general-purpose processor and increase execution time.

This project proposes the design and verification of a RISC-V based System-on-Chip (SoC) for handwritten digit recognition. The SoC integrates a RISC-V processor with instruction memory, data memory, standard peripherals, and four additional hardware IPs: a DMA Controller, CNN Convolution Accelerator, ReLU Unit, and 7-Segment Display Controller. The processor executes the digit-recognition application and controls the hardware accelerators through a memory-mapped bus interface.

The proposed system processes a 28×28 grayscale handwritten digit image, such as an MNIST-style image, and classifies it into one of the ten digits from 0 to 9. Instead of performing the computationally intensive convolution operation entirely in software, the RISC-V processor offloads this operation to a dedicated hardware CNN accelerator. The DMA controller assists in transferring image and CNN data between memory and the accelerator with reduced processor involvement. The recognized digit is displayed on the onboard 7-segment display of the Basys3 FPGA, while UART provides system status and result information.

The project demonstrates the integration of a general-purpose RISC-V processor with application-specific hardware acceleration, memory-mapped peripherals, DMA-based data movement, and FPGA-based output in a compact and modular SoC architecture.

2. Proposed SoC Architecture

The SoC is organized around the RISC-V processor as the central control unit. The processor is connected to instruction memory, data memory, mandatory peripherals, and application-specific accelerator IPs through a common bus interconnect. It executes the application, configures the hardware IPs, sequences the recognition process, and performs the remaining lightweight classification operations.

- Image Storage and Data Movement — A 28×28 grayscale image contains 784 pixel values. The image and CNN weights are stored in Data Memory. The DMA Controller transfers required blocks of data between memory and the accelerator without requiring the processor to copy every element.
- CNN Computation — The CNN Convolution Accelerator performs a simplified 3×3 convolution using pre-trained weights. Dedicated multipliers, adders, and accumulators perform the multiply-accumulate operations in hardware.
- Activation Processing — The ReLU Unit applies $\text{ReLU}(x) = \max(0, x)$, converting negative convolution results to zero while retaining positive values.
- Digit Classification — The processed feature data is returned to the RISC-V, which performs the remaining lightweight processing and selects one of the ten digit classes.
- Output — The predicted digit is written to the 7-Segment Display Controller, which generates the required segment pattern for the Basys3 onboard four-digit display.
- Monitoring — UART provides host communication and debugging information, Timer measures execution time, and GPIO provides general-purpose status/control signals.

The architecture maintains a clear separation of responsibilities: the RISC-V performs system control and software processing, DMA handles data movement, the CNN accelerator performs convolution, ReLU performs activation, and the seven-segment controller provides the physical result display.

3. Selected Additional IPs

- DMA Controller

Transfers blocks of image and CNN data between memory and the accelerator with minimal CPU involvement. It is configured through source address, destination address, transfer length, control, and status registers.

- **CNN Convolution Accelerator**

Performs the computationally intensive 3×3 convolution using dedicated hardware multipliers, adders, and accumulators. It is controlled through memory-mapped registers.

- **ReLU Unit**

Applies the ReLU activation function to convolution results. It is a small hardware block based primarily on comparison and selection logic.

- **7-Segment Display Controller**

Converts the final predicted digit into the appropriate segment pattern and drives/multiplexes the Basys3 onboard four-digit seven-segment display.

4. Mandatory SoC Components

Component	Role
RISC-V Processor	Central processing and system control
Instruction Memory	Stores the application program
Data Memory	Stores image, weights, feature maps, and results
Bus Interconnect	Connects processor, memories, peripherals, and accelerators
UART	Debugging, status, and result communication
Timer	Processing-time and performance measurement
GPIO	General-purpose control and status signals

5. Expected Demonstration

- A 28×28 handwritten digit image is stored in Data Memory.
- The RISC-V configures the DMA and CNN accelerator through memory-mapped registers.
- DMA transfers the required image/weight data.
- The CNN accelerator performs hardware-accelerated 3×3 convolution followed by ReLU processing.
- The RISC-V performs the remaining classification and determines the digit from 0 to 9.
- The predicted digit is displayed on the Basys3 onboard seven-segment display.
- UART reports processing status and recognition results, while the Timer provides execution-time measurements.